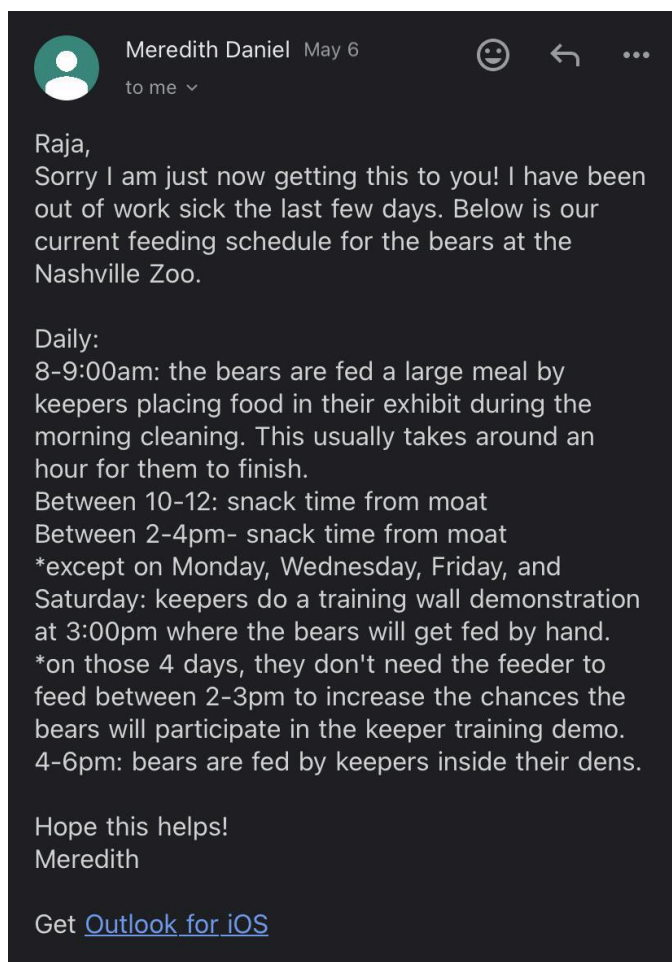


Element J

Raja Swaminathan

Documentation of Evaluation:

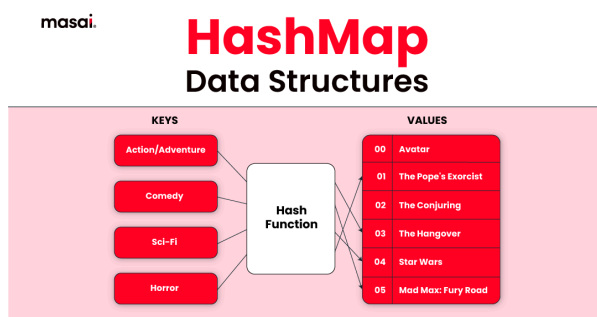
Within our engineering design process, we had two main experts, who had in field experience, who we were able to utilize and get feedback from regarding the specifications of our project itself. The first main person was Meredith Daniel, an in-field expert on the Andean Bears at the Nashville Zoo. Initially, when we started our programming project, we intended for the design to shoot 3 times a day at random times, however with Ms. Daniel's assistance, we were able to deduce that we needed to cater each day of the week to specific times in which the shooter was needed.



The following image is a email sent by Ms. Daniel, the carnivore expert, and in here she explains the specificities of the bear feeding times. This is not only useful because we now were able to cater to each day for the bears, but also we were able to lower the amount of times in which the launching mechanism had to shoot.

```
std::map<int, int*> week;
int sunday[] = {10,12,14,16};
int monday[] = {10,12};
int tuesday[] = {10,12,14,16};
int wednesday[] = {10,12};
int thursday[] = {10,12,14,16};
int friday[] = {10,12};
int saturday[] = {10,12};
week[0] = sunday;
week[1] = monday;
week[2] = tuesday;
week[3] = wednesday;
week[4] = thursday;
week[5] = friday;
week[6] = saturday;
```

The code snippet shown above describes the new plan for dealing with the days Ms. Daniel provided us with.



The code utilizes a hashmap, where the keys are days of the week, represented by integers, and the buckets are arrays with random times within them. Using this we can add specificity to our code.

The other specialist we spoke with was Mr. Bill Rodriguez from Vanderbilt University. He has a specialization in computer science and more specifically sent us an in depth article involving how we can change our motor code.

Link: https://lastminuteengineers.com/l298n-dc-stepper-driver-arduino-tutorial/#google_vignette

With the help of Mr. Rodriguez's Code, we were able to adjust our motor code to something much simpler.

```
void speedControl() {  
  // Turn on motors  
  digitalWrite(in1, LOW);  
  digitalWrite(in2, HIGH);  
  digitalWrite(in3, LOW);  
  digitalWrite(in4, HIGH);  
  // Accelerate from zero to maximum speed  
  for (int i = 0; i < 256; i++) {  
    analogWrite(enA, i);  
    analogWrite(enB, i);  
    delay(20);  
  }  
}
```

The code above shows how to control the speed of the motors, and the next image shows the function of the directional control.

```
void directionControl() {  
  // Set motors to maximum speed  
  // For PWM maximum possible values are 0 to 255
```

```
analogWrite(enA, 255);
analogWrite(enB, 255);
// Turn on motor A & B
digitalWrite(in1, HIGH);
digitalWrite(in2, LOW);
digitalWrite(in3, HIGH);
digitalWrite(in4, LOW);
delay(2000);
// Now change motor directions
/*
digitalWrite(in1, LOW);
digitalWrite(in2, HIGH);
digitalWrite(in3, LOW);
digitalWrite(in4, HIGH);
delay(2000);
*/
// Turn off motors
digitalWrite(in1, LOW);
digitalWrite(in2, LOW);
digitalWrite(in3, LOW);
digitalWrite(in4, LOW);
}
```

With the help of these two specialists, we were specifically able to tailor our code to work better with not just design the launching build team had created, but also whatever the zoo was in search of as well.